

TOOLARA, Australia

WMO: 945580

Lat: 25.971S

Lon: 152.861E

Elev: 143

StdP: 99.62

Time Zone: 10.00 (E10)

Period: 01-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	10.0	-2.1	3.2	15.7	-0.2	3.8	15.5	7.9	16.1	7.0	14.3	4.4	240	0.349

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.4	33.2	23.2	31.7	22.8	30.5	22.3	25.2	30.0	24.5	28.8	23.9	27.8	4.5	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.0	19.2	26.8	23.4	18.5	26.2	22.8	17.9	25.7	77.6	29.9	75.0	28.6	72.5	27.8	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	7.5	6.4	DB	6.7	36.8	1.2	1.5	5.8	37.8	5.1	38.7	4.4	39.5	3.5	40.6	(4)
(5)			WB	3.5	26.8	1.0	0.7	2.8	27.3	2.2	27.7	1.6	28.0	0.8	28.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.1	25.2	25.0	23.5	21.7	18.5	16.5	16.0	17.3	19.9	21.9	23.3	24.7	(6)	
	DBStd	3.83	2.00	1.84	1.70	1.68	1.83	2.17	1.77	2.06	2.33	2.35	1.98	2.39	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	215	0	0	0	1	20	62	75	46	11	2	0	0	(9)	
	CDD10.0	4050	471	421	419	349	263	195	186	226	297	368	399	456	(10)	
	CDD18.3	1223	212	188	161	100	25	7	2	13	57	111	149	198	(11)	
	CDH23.3	6859	1416	1208	756	334	49	6	3	46	275	593	845	1327	(12)	
(13)	CDH26.7	2057	486	393	181	42	2	0	0	4	56	163	236	494	(13)	
(14)	Wind	WSAvg	3.4	3.8	3.6	3.7	3.4	3.4	3.3	3.2	3.2	3.5	3.4	3.5	(14)	
(15) Precipitation	PrecAvg	1288	186	205	166	102	100	89	54	71	48	73	80	151	(15)	
	PrecMax	1910	447	507	480	250	304	221	171	442	178	177	154	489	(16)	
	PrecMin	863	18	27	42	33	26	17	12	0	1	17	8	55	(17)	
	PrecStd	282	118	145	116	50	63	58	39	105	48	46	39	103	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.9	34.7	32.2	29.6	26.3	24.3	23.6	26.7	30.6	32.7	32.8	35.6	(19)	
		MCWB	24.6	24.4	23.6	21.4	18.3	18.8	15.5	17.7	18.2	20.0	21.9	23.7	(20)	
	2%	DB	32.5	31.9	30.1	27.8	24.7	22.6	22.1	24.5	28.2	30.1	30.8	33.0	(21)	
		MCWB	23.7	24.0	22.4	20.6	17.3	17.4	15.3	16.3	18.5	19.7	21.6	23.2	(22)	
	5%	DB	30.9	30.4	28.6	26.5	23.4	21.3	20.9	22.9	26.4	28.4	29.3	31.2	(23)	
		MCWB	23.2	23.4	21.7	20.2	17.2	16.7	15.0	15.5	17.7	19.5	21.0	22.6	(24)	
	10%	DB	29.5	29.1	27.2	25.3	22.1	20.1	19.7	21.5	24.7	26.7	27.8	29.5	(25)	
		MCWB	22.6	22.8	21.4	20.0	17.0	16.2	14.9	14.8	17.1	18.9	20.3	22.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.0	26.4	25.2	23.7	20.2	20.2	19.7	19.8	21.2	22.9	23.9	25.7	(27)	
		MCDB	31.8	32.1	29.3	26.5	23.2	22.1	21.4	23.7	26.1	29.1	29.8	32.0	(28)	
	2%	WB	24.9	25.2	24.1	22.5	19.5	19.1	18.3	18.3	20.1	21.8	23.0	24.7	(29)	
		MCDB	29.6	29.8	27.3	25.0	22.4	20.9	19.6	21.7	25.0	26.8	27.9	29.6	(30)	
	5%	WB	24.2	24.4	23.3	21.6	18.7	18.3	16.9	17.4	19.2	21.0	22.2	23.9	(31)	
		MCDB	28.3	28.5	26.2	24.5	21.4	20.0	19.0	20.4	23.4	25.3	26.7	28.1	(32)	
	10%	WB	23.7	23.7	22.5	20.8	18.1	17.5	15.9	16.4	18.5	20.2	21.4	23.1	(33)	
		MCDB	27.4	27.4	25.4	23.9	20.5	19.2	18.6	19.5	22.3	24.4	25.7	27.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	8.3	7.7	8.1	8.0	7.4	8.5	9.6	10.1	10.2	9.5	9.4	(35)	
		MCDBR	10.7	10.3	9.9	9.6	9.4	8.2	9.4	10.8	12.2	12.5	11.6	12.2	(36)	
	5% WB	MCWBR	3.5	3.6	3.7	3.9	4.4	4.5	4.8	4.5	4.5	4.5	4.0	4.3	(37)	
		MCWBR	8.9	9.0	7.8	7.9	7.3	6.4	7.0	9.1	9.9	10.8	10.1	10.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.384	0.377	0.359	0.336	0.318	0.312	0.301	0.312	0.351	0.370	0.373	0.387	(40)	
		taud	2.519	2.545	2.592	2.642	2.640	2.652	2.667	2.617	2.497	2.476	2.504	2.493	(41)	
	Ebn at Noon		959	950	939	918	894	879	904	929	932	946	963	958	(42)	
		Edh at Noon	113	108	98	85	78	73	75	86	106	114	114	117	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.56	5.98	5.20	4.54	3.79	3.22	3.69	4.58	5.61	6.17	6.70	6.68	(44)	
	RadStd		0.53	0.52	0.39	0.31	0.20	0.24	0.32	0.36	0.43	0.52	0.61	0.73	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (21 neighbors)	+0.37	+0.56	N/A	N/A	N/A	N/A	N/A	-70	+183	+110	

Nomenclature: See separate page